BRIAN MORIARTY · SRECON26 EMEA, DUBLIN · OCTOBER 2026

Are You Talking to Me?

Confidence entanglement in AI-assisted operations

An AI assistant now sits in the path of incident diagnosis, in the seat a second reviewer would hold. That arrangement produces a failure I am calling confidence entanglement. The assistant occupies the procedural position of an independent check without the separation that position needs, so what it returns reaches the responder as confirmation rather than as review. After a few exchanges the responder can no longer say which half of the diagnosis is theirs, which is where the term comes from: two systems are entangled when there is no longer any way to describe them separately. Every piece of that is familiar, since it rides on confirmation bias, collects the deference people already give automation, and counts a restatement of the responder's own account as fresh support. What is new is where those land. The review defenses a team already has, the second pair of eyes and the challenge in the channel, were built for a weak reviewer, somebody who can be argued with, and they are not aimed at a response assembled out of the responder's own account.

One mechanism I am proposing is register. When an assistant answers in the voice of a reviewer, the responder may grant it a reviewer's authority. An organized answer in that voice can suppress the moment the responder would otherwise have stopped to look for alternatives, because it reads as though the alternatives were already weighed. Turpin and colleagues found that a model's stated reasoning can follow features of the prompt rather than the reasons it acted on, so an account that reads as deliberation is not evidence that any deliberation happened. The working reflex shifts from testing a theory to executing a diagnosis, and nothing in the exchange calls for the alternative the responder never mentioned.

The underlying effects are familiar. Google's 2026 SRE volume names multiple hypotheses as its preferred investigation method, because the framing "reduces confirmation bias by asking responders to disprove ideas rather than confirm them". The same chapter's guidance on AI then has the agent "generate a high-confidence incident hypothesis" and hand the on-caller "a single credible lead",

and offers it as a “rubber duck” against which a responder checks assumptions. A single lead may help with triage; the concern is what happens when it reaches a consequential decision without challenge.

Separation, not independence. Independence is a matter of degree rather than a property somebody has. A dependency owner who has read the channel is already exposed to the framing and can still bring evidence the responder did not select. The assistant is exposed in a further way, because the account it was handed shapes what evidence it goes and gets. A live incident rarely supplies anyone who read none of it, and Google’s own practice places the independent reviewer in the postmortem, chosen because they did not take part. What remains is a question of how much separation, not whether. A team can build separation, which is a procedure rather than a person, and should not call it independence. Four conditions: Ask for rival causes adversarially. Write down the signal that would tell two of them apart, before running the query that would settle it. Check that the two causes predict different values for it. Decide in advance what a contrary result would change, and change it.

A worked example. Consider a latency spike. A responder who has seen something similar enters the reading into the assistant, that it resembles a prior cache-eviction problem. The assistant returns an organized account of why eviction fits. The team raises the cache memory limit, latency improves for a few minutes, and the incident continues. The partial improvement reads as confirmation, because the panel in front of them shows backend execution time rather than the time a request waits for a connection. So they go further and flush the cache, to clear what they now believe is a thrashing working set. Nothing restores the entries that were in it. The cold cache sends more traffic downstream, into a connection pool already starving under a retry storm, and the incident gets worse. Pool saturation was never raised as an alternative. The assistant could have raised it. Nothing in the loop asked it to look past the framing it was handed.

Run the same incident with the form in front of the flush. The responder asks for the strongest rival causes and the signal that would separate each from the current reading, and writes it down before running the query. The result is not clean. A bad deploy is weakened by there having been no deploy in the window, without being ruled out. For another rival the separating signal is a metric the team does not emit. One remains checkable, pool saturation, named before anyone looked, with the prediction written as utilization at or above 95 percent for five minutes and eviction at baseline. Utilization then crosses the line and holds for four minutes rather than five, so the prediction comes in short of what

was written. That does not rule out pool saturation, but it does not meet the pre-committed test. The flush does not happen. The action taken instead is to cap retries and shed optional traffic, which reduces load whether the pool or the cache is at fault, with the rollback trigger set before anyone pressed anything.

From advice to workflow. The pre-commitment above is workflow design rather than individual skill. Under load, composing a careful prompt and keeping a hypothesis record is the first thing to go while a responder is coordinating others and weighing a rollback. Optional steps are dropped first, and putting it in a wiki as recommended practice will not work. It belongs in the incident channel as a form, required before a defined class of hard-to-reverse actions, with a named owner and the evidence at decision time captured automatically. Its fields are the current read, the strongest rival, the signal that separates them with the value and window expected, the proposed action, and what would trigger a rollback. One line is most meaningful: The responder names what they expect to see if the reading is right, with a threshold and a window, before running the query chosen to test it. The bot fills in the context, the service, the window and the metric names it can see, while the read, the rival and the prediction are the responder's, because a challenge the assistant wrote is not by itself a challenge. A break-glass path keeps the form from blocking action when delay is the greater danger, and the incident commander may proceed at once, with reason and time recorded and a retrospective required.

What a team can do. Take one class of hard-to-reverse action, place the form in front of it in the incident channel, and enforce the single line. Draw that class by how hard recovery is rather than by the name of the command, because a rollback can be the fastest way back or the dangerous move after a schema change. Hand a separate responder the raw signals and withhold the diagnosis, so their read forms before they see anyone else's. Ask whether separation is something the team considers, and check the metric as well, because an agent scored on the share of its suggestions that people accept is scored on agreement.

What this brief does not have. No study sits behind the arrangement. The component effect has been measured, in that AI explanations raise the rate at which people accept a recommendation whether or not it is correct, and over-reliance on automation has its own literature. I know of no direct test of the arrangement, there is no test of whether register is the cause, and there is no evidence that the form improves how an incident comes out. The argument is from structure, and an argument from structure can show where a failure has room to happen without showing how often it happens.

The form costs review time, and it can fail in the same way everything else here fails, by being filled in and counted as a review that nobody performed. Switch it off if break glass becomes the normal path, if time to a safe mitigation rises against your own baseline, or if the written prediction is never scored against what came back.

Measuring it is the next piece of work, and I am setting that up now. It needs incident responders willing to spend twenty minutes on a vignette, and a team willing to run the gate in one channel for a quarter. Even an off-the-record conversation about how your channel actually runs is useful.
brian.moriarty@stevens.edu

Sources. *Site Reliability Engineering*, 2nd ed., eds. Beyer, Jones, Leng, Huska, Petoff and Murphy (O'Reilly, 2026), chapters 10, 11 and 21. Gagan Bansal et al., "Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance", CHI 2021. Miles Turpin, Julian Michael, Ethan Perez and Samuel R. Bowman, "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting", NeurIPS 2023. Benjamin Enke and Florian Zimmermann, "Correlation Neglect in Belief Formation", *Review of Economic Studies* 86 (2019): 313-332. Raja Parasuraman and Victor Riley, "Humans and Automation: Use, Misuse, Disuse, Abuse", *Human Factors* 39, no. 2 (1997): 230-253. Kate Goddard, Abdul Roudsari and Jeremy C. Wyatt, "Automation Bias: A Systematic Review of Frequency, Effect Mediators, and Mitigators", *JAMIA* 19, no. 1 (2012): 121-127.